

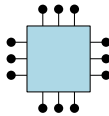
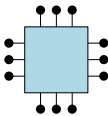
Age-Based Scheduling for a Memory-Constrained Quantum Switch

Stavros Mitrolaris Subhankar Banerjee Sennur Ulukus

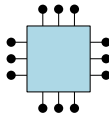
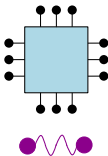
University of Maryland, College Park, MD.

IEEE INFOCOM (QuNAP) 2026

Direct Entanglement



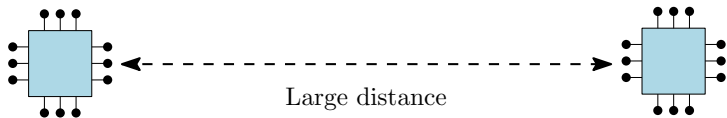
Direct Entanglement



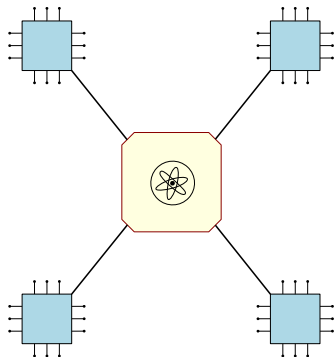
Direct Entanglement



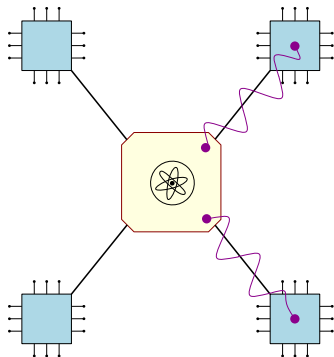
Direct Entanglement



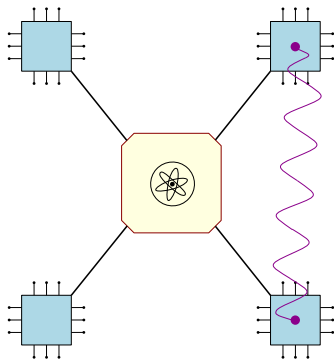
Quantum Switch



Quantum Switch



Quantum Switch



The Scheduling Challenge

- ▶ Quantum memory registers are a **scarce resource**.
- ▶ Hard memory constraint of M registers.
- ▶ Closest system model: Promponas et al.¹
 - ▶ Hard memory constraint.
 - ▶ Throughput optimal policy.
 - ▶ **But:** exponentially many NP-hard problems per slot.
- ▶ **Question:** What can we do differently?

¹P. Promponas, V. Valls, S. Guha, and L. Tassiulas. Maximizing entanglement rates via efficient memory management in flexible quantum switches. IEEE JSAC, March 2024

Main Objectives

Design scheduling policies that:

- ▶ Account for hard memory constraints.
- ▶ Implementable.
- ▶ Provide QoS guarantees.

Need: novel **QoS-oriented metric** for performance evaluation.

Beyond Throughput

Optimizing throughput:

- ▶ Maximizes the **total** number of requests served per slot.
- ▶ No **per-request** guarantees.
- ▶ Can have requests starving.
- ▶ Does not easily translate to QoS guarantees per request.

We introduce the **Age of Entanglement Establishment**.

Beyond Throughput

Key Observation: Think of serving a request as serving a job².

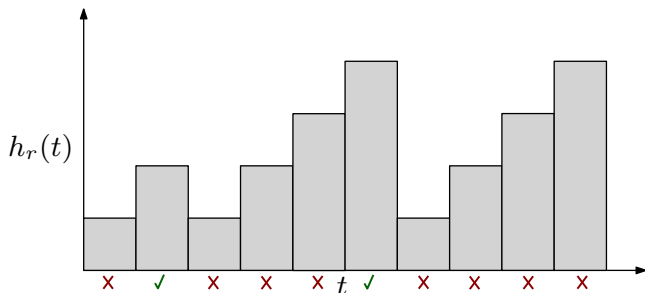
Age of entanglement establishment (AoEE):

$$h_r(t) = t - \sup\{\tau \in \{1, 2, \dots, t-1\} : d_r(\tau) = 1\}.$$

For request r , how long has it been since it last got served?

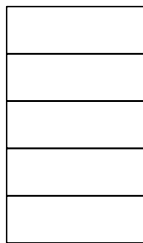
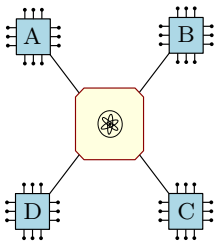
²S. Mitroliaris, S. Banerjee, and S. Ulukus. Age of job completion minimization with stable queues. In IEEE ICC, May 2026

AoEE Example



- ▶ Throughput: "The switch serves 5 requests per slot".
- ▶ AoEE: "On average, request r gets served every 3 slots".

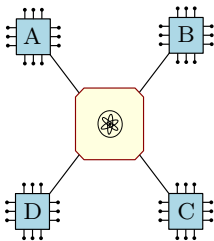
System Model Example



$$M = 5$$

► $\mathcal{R} = \{AB, CD, ABC, BCD, ABCD\}$.

System Model Example

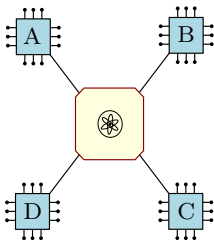


C
D
A
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$$M = 5$$

► $\mathcal{R} = \{AB, \textcolor{red}{CD}, \textcolor{blue}{ABC}, BCD, ABCD\}.$

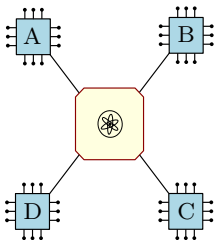
System Model Example



$M = 5$

- ▶ $\mathcal{R} = \{AB, \textcolor{brown}{CD}, \textcolor{blue}{ABC}, BCD, ABCD\}$.
- ▶ Probabilistic LLEs.
- ▶ Probabilistic swapping.

System Model Example



$$M = 5$$

- ▶ $\mathcal{R} = \{AB, \textcolor{red}{CD}, \textcolor{blue}{ABC}, BCD, ABCD\}$.
- ▶ Assumptions:
 - ▶ Full load.
 - ▶ One-slot decoherence.
 - ▶ No LLE repurposing.

The Combinatorial Blowup

- ▶ Cardinality of a request = the # users participating.
- ▶ Can have all requests of cardinality $2, 3, \dots, N$.
 - ▶ $|\mathcal{R}| = 2^N - (N + 1)$.
- ▶ Memory configurations: **Combinatorial** in $|\mathcal{R}|$ and M .
- ▶ Vast action space.

Our Approach:

- ▶ **Structured** policies: complexity **linear** in $|\mathcal{R}|$.
- ▶ **Closed-form** AoEE guarantees.

Single-Cardinality Stationary Randomized (SSR)

Key observation: Restricting to requests of a **single cardinality** λ eliminates the combinatorial problem.

Step 1

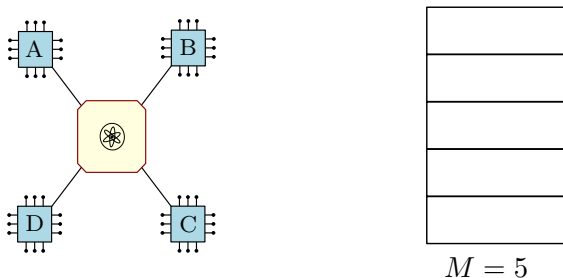
Sample cardinality a λ
from $\Lambda = \{\lambda_1, \lambda_2, \dots, \lambda_k\}$



Step 2

Sample as many requests of cardinality λ
as possible, without replacement.

Single-Cardinality Stationary Randomized (SSR)



$$\mathcal{R} = \{\text{AB}, \text{CD}, \text{ABC}, \text{BCD}, \text{ABCD}\}$$

- 1) Prob. distr. over $\Lambda = \{2, 3, 4\}$, say we sample $\lambda = 3$.
- 2) Sample one request from $\{\text{ABC}, \text{BCD}\}$.

Single-Cardinality Stationary Randomized (SSR)

Lemma 1

For any policy $\mu \in \Pi_{\text{SSR}}$, the average age is given by

$$\Delta^\mu = \frac{1}{R} \sum_{\lambda \in \Lambda} \frac{1}{\mu^{(0)}(\lambda) q_\lambda} \sum_{r \in \mathcal{C}(\lambda)} \frac{1}{\mu^{(\lambda)}(r) v(r)}.$$

Single-Cardinality Stationary Randomized (SSR)

Theorem 1

Theorem

Let $\mu^* = (\mu^{*(0)}, \mu^{*(\lambda_1)}, \mu^{*(\lambda_2)}, \dots, \mu^{*(\lambda_k)})$ denote the optimal SSR policy. Then, for each $\lambda \in \Lambda$ and $r \in \mathcal{C}(\lambda)$,

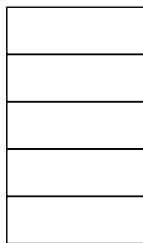
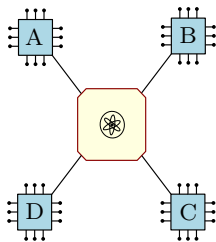
$$\mu^{*(\lambda)}(r) = \begin{cases} \min \left\{ 1, \frac{1}{\sqrt{\gamma^{(\lambda)} v(r)}} \right\}, & \text{if } M_\lambda < |\mathcal{C}(\lambda)| \\ 1, & \text{if } M_\lambda = |\mathcal{C}(\lambda)|, \end{cases}$$

and,

$$\mu^{*(0)}(\lambda) = \frac{\sqrt{f_\lambda^*/q_\lambda}}{\sum_{\lambda' \in \Lambda} \sqrt{f_{\lambda'}^*/q_{\lambda'}}},$$

where $f_\lambda^* := \sum_{r \in \mathcal{C}(\lambda)} \frac{1}{\mu^{*(\lambda)}(r)v(r)}$.

Single-Cardinality Max-Weight (SMW)



$$M = 5$$

$$\mathcal{R} = \{\text{AB}, \text{CD}, \text{ABC}, \text{BCD}, \text{ABCD}\}$$

- 1) Prob. distr. over $\Lambda = \{2, 3, 4\}$, say we sample $\lambda = 3$.
- 2) Schedule the request with **max weight** from $\{\text{ABC}, \text{BCD}\}$.

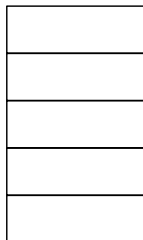
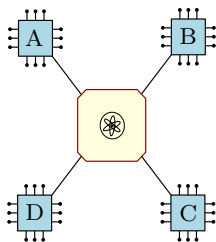
Single-Cardinality Max-Weight (SMW)

Theorem 2

The SMW policy ψ performs no worse than the optimal SSR policy μ^* ,

$$\Delta^\psi \leq \Delta^{\mu^*}.$$

Multi-Cardinality Max-Age (MMA)



$$M = 5$$

$$\mathcal{R} = \{\text{AB}, \text{CD}, \text{ABC}, \text{BCD}, \text{ABCD}\}, \Lambda = \{2, 3, 4\}$$

$$\Lambda^{(1)} = \{2, 3\}, \Lambda^{(2)} = \{4\}.$$

- 1) With some prob. distr. choose between $\Lambda^{(1)}$ and $\Lambda^{(2)}$, say we choose $\Lambda^{(1)}$.
- 2) Schedule **max age** request in $\{\text{AB}, \text{CD}\}$ **and** $\{\text{ABC}, \text{BCD}\}$.

Multi-Cardinality Max-Age (MMA)

Key observation: Selecting a predetermined cardinality subset $\Lambda^{(i)} \subseteq \Lambda$ eliminates the combinatorial problem.

Step 1

For a predetermined collection of cardinality subsets $\{\Lambda^{(i)}\}_{i \in [n]}$, sample one Λ_i .



Step 2

For each cardinality $\lambda \in \Lambda^{(i)}$, schedule the request of cardinality λ with max age.

Multi-Cardinality Max-Age (MMA)

Lemma 2

For any policy $\phi \in \Pi_{\text{MMA}}(\{\Lambda^{(i)}\}_{i \in [n]})$, the average age of a request $r \in \mathcal{C}(\lambda)$ is given by,

$$\Delta_r^\phi = \frac{1}{2\theta_\lambda q_\lambda} \left(\frac{1}{\beta_\lambda} \sum_{r' \in \mathcal{C}(\lambda)} \frac{1}{v(r')^2} + \beta_\lambda \right),$$

where $\theta_\lambda := \sum_{i: \lambda \in \Lambda^{(i)}} \phi(i)$ and $\beta_\lambda := \sum_{r' \in \mathcal{C}(\lambda)} \frac{1}{v(r')}$.

- Requests of same cardinality have the same average age.

Multi-Cardinality Max-Age (MMA)

Proposition 1

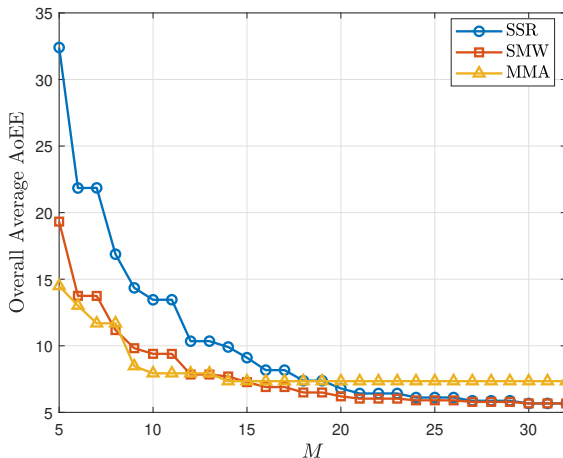
The optimal policy $\phi^* \in \Pi_{\text{MMA}}(\{\Lambda^{(i)}\}_{i \in [n]})$ employs the sampling distribution ϕ^* given by,

$$\begin{aligned} \phi^* \in \arg \min_{\phi} \quad & \sum_{\lambda \in \Lambda} \frac{1}{\sum_{i: \lambda \in \Lambda^{(i)}} \phi(i)} w_{\lambda} \\ \text{s.t.} \quad & \phi(j) \geq 0, \quad \sum_{j \in [n]} \phi(j) = 1, \end{aligned}$$

where $w_{\lambda} := \frac{|\mathcal{C}(\lambda)|}{2q_{\lambda}} \left(\frac{1}{\beta_{\lambda}} \sum_{r \in \mathcal{C}(\lambda)} \frac{1}{v(r)^2} + \beta_{\lambda} \right)$.

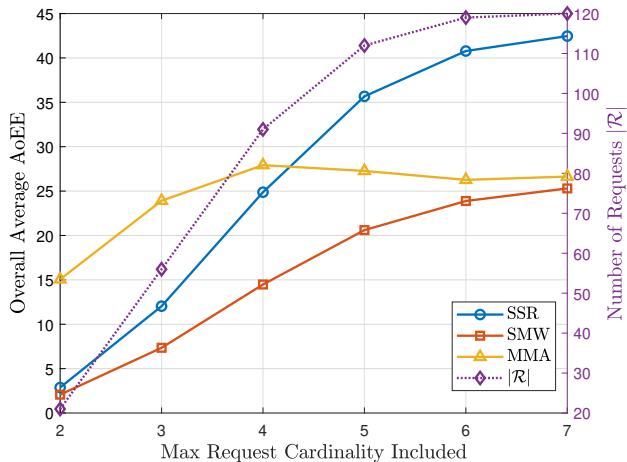
- **Convex** optimization problem that can be solved **once offline**.

Numerical Analysis: Effect of Memory



$N = 5$ users and all possible requests (26).

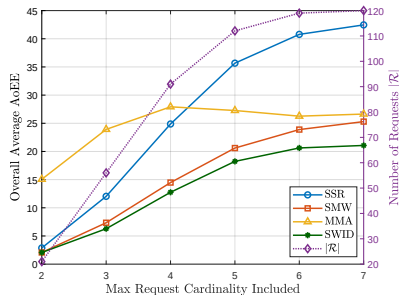
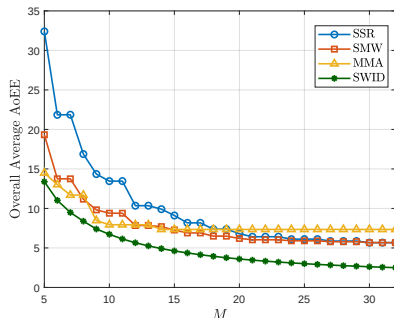
Numerical Analysis: Effect of Request Set



$N = 7$ users and $M = 20$ memory registers.

Index-Based Policies

Formulate in **Restless Multiarmed Bandit** framework³.



³S. Banerjee, S. Mitrolaris, and S. Ulukus. Index-based scheduling for a resource-constrained quantum switch. IEEE WiOpt, June 2026.

Conclusions

- ▶ Age-based metric for quantum networks.
- ▶ Linear complexity scheduling algorithms with AoEE guarantees.
- ▶ SMW always outperforms SSR.
- ▶ MMA: Better for small memory (multiple cardinality requests).
- ▶ SMW: More robust.
- ▶ Index-based greedy: Outperforms SSR, SMW and MMA.